Dataset Source

The dataset used in this project is the Brazilian E-Commerce Public Dataset by Olist, which contains information about orders, customers, products, payments, sellers, and customer reviews from an online marketplace.

The dataset consists of multiple tables that were cleaned and prepared for sales analysis.

1. **Orders Dataset:** File: olist_orders_dataset.csv

Description**:** This dataset contains information about customer orders including order status, purchase timestamps, delivery dates, and estimated delivery times.

Cleaning Steps Performed

1. Missing Value Analysis: Missing values were identified in the following columns:

[order_ approved_ at, order_ delivered_ carrier_ date, order_ delivered_ customer_ date]

These missing values were retained because they represent valid business situations such as orders that were not yet approved, shipped, or delivered.

2. Duplicate Check: Duplicate records were checked using:

orders. duplicated().sum(). No duplicate rows were found.

3. Date Conversion: Timestamp columns were converted into datetime format for time-based analysis. Converted columns:

[order_ purchase_ timestamp, order_ approved_ at**,** order_ delivered_ carrier_ date**,** order_ delivered_ customer_ date, order_ estimated_ delivery_ date

4. Feature Engineering: Additional columns were created to support time-based analysis**.**

Created columns: order_ year, order_ month, delivery_ days

2. Customers Dataset: File: olist_customers_dataset.csv

Description: This dataset contains customer information including location and unique customer identifiers. Cleaning Steps Performed

1. **Missing Value Check**: No missing values were found.

2. Duplicate Check: Duplicate records were checked using, **"**customers. duplicated().sum()",No duplicate records were found.

3. Data Standardization: City names were standardized using title formatting to ensure consistency in geographic analysis. Example transformation: saopaulo → Sao Paulo

3. Order Items Dataset: File: olist_order_items_dataset.csv

Description: This dataset contains detailed information about products included in each order.

Cleaning Steps Performed:

1. Missing Value Check: No missing values were found.

2. Duplicate Validation: Duplicates were checked using different combinations of columns.

Checking duplicates using only order_ id showed repeated values because a single order can contain multiple products. The dataset was validated using the unique combination:

order_ id + order_ item_ id: No duplicate records were found using this unique identifier.

3. Revenue Feature Creation: A new column called total_ price was created to calculate total product cost including shipping. Formula used:

total_ price = price + freight_ value

4. Products Dataset: File: olist_products_dataset.csv

Description**:** This dataset contains information about product categories and physical product attributes.

Cleaning Steps Performed

1. Missing Value Handling: Missing values were found in several columns.

- product_ category_ name contained 610 missing values. These were replaced with "Unknown".
- product_ name_ length, product_ description_ length, and product_ photos_ qty also contained 610 missing values and were replaced with 0.

2. Product Dimension Handling: Columns with missing values [product_ weight_ g, product_ length_ cm**,** product_ height_ cm, product_ width_ cm]

These values were replaced with the median value of each column to maintain realistic product measurements.

3. Duplicate Check: Duplicate records were checked using: products. duplicated().sum()

No duplicates were found.

5. Payments DatasetFile: olist_order_payments_dataset.csv

Description: This dataset contains payment transaction information including payment type, installments, and payment value. Cleaning Steps Performed

1. Missing Value Check: No missing values were detected.

2. Payment Value Validation: Payment values were checked to ensure all transactions contained valid amounts. Some records contained payment_value = 0. These records correspond to orders fully paid using vouchers or discount credits, so they were retained.

3. Duplicate Check: Duplicate records were checked using: payments. duplicated().sum()

No duplicates were found.

6. Reviews Dataset: File: olist_order_reviews_dataset.csv

Description**:** This dataset contains customer feedback including review scores and comments. Cleaning Steps Performed

1. Missing Value Handling**:** Missing values were found in: [review_ comment_ title, review_ comment_ message]. These occur because many customers provide only ratings without comments. Missing values were replaced with, No Comment

2. Review Score Validation: Review scores were validated to ensure they fall within the expected rating range. The unique values found were [1, 2, 3, 4, 5], This confirms valid rating data

3. Date Conversion: Converted the following columns to datetime format:

[review_ creation_ date, review_ answer_ timestamp]

4. Duplicate Check: No duplicate records were found.

5. Output: Cleaned dataset saved as: reviews_cleaned.csv (according to data set name)